B.Tech IT/ CSE 3rd Sem

Mid Term Examination October 2022

Paper Code: BIT 203 **Paper: Software Engineering**

Note: i) All Questions are compulsory. Internal choices have been provided.

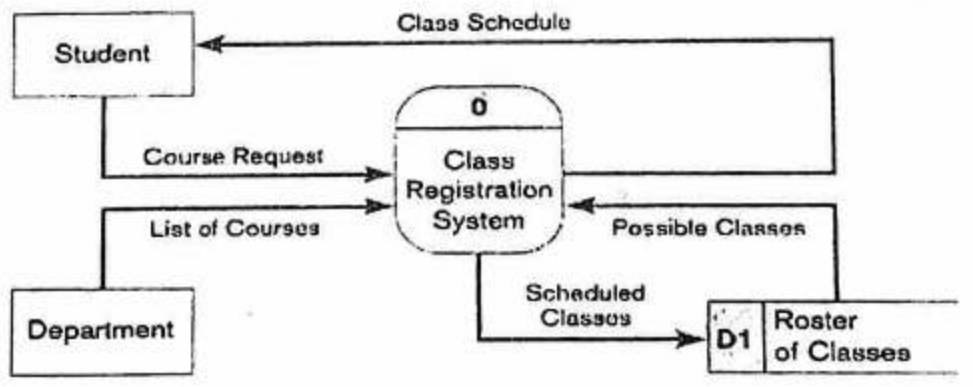
ii) Attempt all parts of the question together? Start each question from fresh page.

Q1		
a)	Requirement elicitation aims at reconciling domain knowledge requirements and use-case requirements. Explain the difference between these two kinds of requirements. Should one of them take precedence over the other in the requirements determination process? Give Reasons?	5
	OR	
a)	(i) What is difference between a survey and a questionnaire in requirements determination and similar tasks? (ii) What is scope creep? How it can be addressed during requirements determination?	3, 2
Q2	Do you think quantitative project data is important. In what way can it contribute to project planning?	5
	OR	
	What are the differences between good requirements and a good requirements specification?	5
Q3		
a)	(i) Define the terms: Measure, Measurement and metric mean? (ii) What are software configuration items?	3,2
	OR	
a)	(i) Why is requirement traceability important? (ii) Why should requirements be numbered?	3, 2
Q4	Do you think the linear model of software development is appropriate? In which cases do you think an agile approach is more appropriate?	5
	OR	
	Suppose you are involved in a large project concerning the development of a patient planning system for a hospital. You may opt for one of two strategies. The first strategy is to start with a thorough analysis of user requirements, after which the system is build according to these requirements. The second strategy starts with a less complete requirements analysis phase, after which a pilot version is developed. The pilot version is installed in a few small departments. Further development of the system is guided by the experience gained in working with the pilot version. Discuss the pros and cons of both strategies. Which strategy do you favour?	5
Q5	Develop a context diagram to represent necessary details of the training logistics system described in the following narrative. Training Inc. conducts training seminars in major cities. For each seminar, the logistics department must make arrangements for the meeting facilities, the training consultant's travel and the shipment of any seminar material. For each scheduled seminar, the bookings department notifies the logistics co-ordinator of the type of seminar, the dates and the city locations, and the name of the consultant who will conduct the training. To arrange for meeting facilities, the logistics co-ordinator gathers information on possible meeting sites in the scheduled city. The meeting site location decision is made based on the date availability, cost, type of meeting space available, and the convenience of the location. Once the site decision is made, the co-ordinator speaks with the sales manager of the meeting facility to reserve the meeting room(s), plan the seating arrangement(s),	5

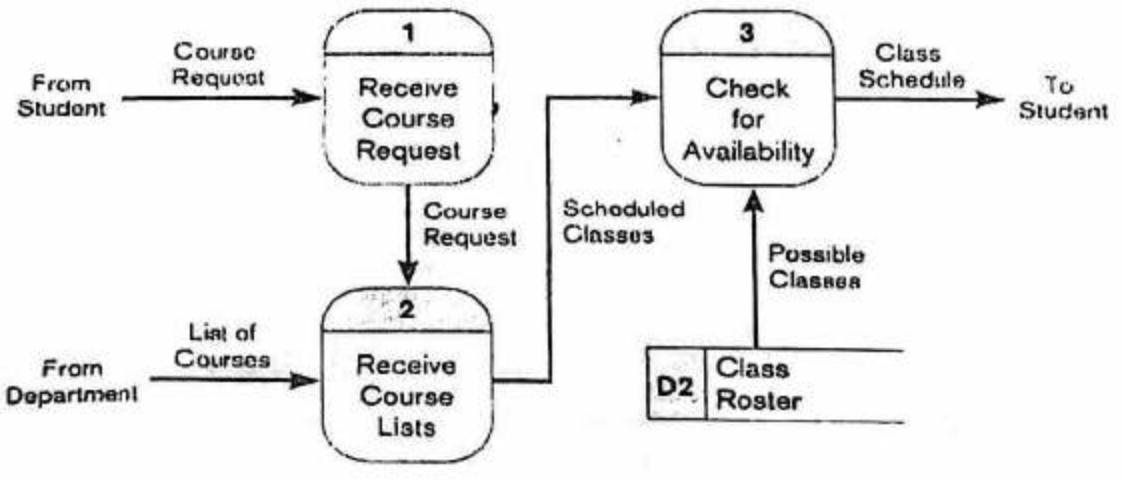
and reserve necessary audio-visual equipment. The co-ordinator estimates the number and size of meeting rooms, type of seating arrangement and the audio-visual equipment needed for each seminar from the information kept in the logistics database on each type of seminar offered and the number of anticipated registrants for a particular booking. After negotiations are conducted by the logistics co-ordinator and the sales manager of the meeting facility, the sales manager creates a contract agreement specifying the negotiated arrangements and sends two copies of it to the logistics co-ordinator. The co-ordinator review the agreement and approves it if no changes are needed. One copy of the agreement is filed and the other copy is sent back to the sales manager. If changes are needed, the agreement copies are changed and returned to the sales manager for approval. This approval process continues until both parties have approved the agreement. The co-ordinator must also contact the training consultant to make the travel arrangement. First, the co-ordinator, reviews the consultants travel information in the logistics database and researches flight schedules. Then the consultant is contacted to discuss the possible travel arrangements; subsequently, the co-ordinator books a flight for the consultant with a travel agency. Once the consultant's travel arrangements have been completed, a written confirmation and itinerary are sent to the consultant. Two weeks before the date of the seminar, the coordinator determines what, if any, seminar materials (e.g., transparencies, training guides, pamphlets, etc.) need to be sent to the meeting facility. Each type of seminar has a specific set of materials assigned to it. For some materials, the coordinator must know how many participants have registered for the seminar in order to determine how many to send. A request for materials is sent to the Materials-handling department, where the materials are gathered, boxed, and sent to the meeting address listed on the request. Once the requested materials have been shipped, a notification is sent to the logistics coordinator.

Q6	Figure below contains draft of context and level-1 DFD for a University class registration system. Identify and explain potential violetions of rules and guidelines on these diagrams. Figure below contains draft of context and level-1 DFD for a Unviersity class registration system. Identify and explain potential violetions of rules and guidelines on these diagrams.	5

Context Diagram



Level-1 Diagram



560